Binary Cataract Detection from ODIR Fundus Images

A comparison of a custom CNN, ResNet18 and EfficientNet-B0 with validation-based threshold selection, patient-level bootstrap intervals and Grad-CAM checks

Abstract

Cataract is a leading cause of blindness, and colour fundus photographs are a possible route to low-cost screening. This project builds and evaluates binary cataract detectors on eye images from the Ocular Disease Intelligent Recognition (ODIR) dataset. After eligibility filtering, 6,500 images from 3,345 patients (287 cataract, 4.4%) were divided by patient into training, validation and held-out test sets. A custom convolutional network, an ImageNet-pretrained ResNet18 and an ImageNet-pretrained EfficientNet-B0 were compared. Checkpoints and the primary model were selected on validation average precision (AP), and two operating policies were kept separate: a common reference threshold of 0.5, and model-specific validation-selected thresholds that target at least 90% validation sensitivity.

On the held-out test set (973 images, 41 cataract), AP and ROC-AUC were 0.653 and 0.945 for the custom CNN, 0.915 and 0.993 for ResNet18, and 0.908 and 0.992 for EfficientNet-B0. EfficientNet-B0, selected on validation, is the primary model. ResNet18 had the highest observed test scores, but patient-level bootstrap intervals for the difference included zero, so neither superiority nor equivalence was established. At the reference threshold both pretrained models detected 38 of 41 cataract images, with 17 and 22 false positives. At the validation-selected thresholds, EfficientNet-B0 was the only main model to reach 90% sensitivity on the held-out test set (92.68%). An exploratory matched-cohort experiment found no clear ranking difference between ResNet18 trained with healthy-only negatives and with mixed negatives, and its threshold-dependent differences changed with the operating policy. A Grad-CAM parameter-randomization check was inconclusive. Limitations include 41 test-set cataract images, no external validation, uncleared near duplicates across patients and missing per-image prediction files.

Keywords: cataract detection; fundus photography; ODIR; transfer learning; threshold selection; bootstrap; Grad-CAM

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1 Introduction

1.1 Background and Motivation

Cataract is a leading cause of blindness: in 2020 it accounted for an estimated 15.2 million cases of blindness among adults aged 50 years and older [1]. Clinical assessment and grading rely on specialist examination of the lens, which limits large-scale screening [2]. Colour fundus photographs are already collected in many retinal screening settings, and lens opacity reduces their clarity [2]. Deep learning systems have therefore been proposed to detect or grade cataract from fundus photographs [2]–[4].

Published results are hard to compare, because cohorts, label definitions, class balance and evaluation protocols differ. In population-based data the prevalence of visually significant cataract by eye can be low; one study reported 1.04% to 6.05% across its test sets [3]. At such prevalence, precision, false-positive burden and threshold choice matter as much as headline accuracy [11]. Two further design issues are easy to overlook. Both eyes of a patient are correlated, so images from one patient must stay in one data split [2]. The non-cataract class also mixes healthy eyes with eyes that have other diseases, so measured performance can depend on the case mix [14].

1.2 Problem Statement and Scope

This report studies binary detection of cataract (versus no recorded cataract) from single colour fundus images in the public ODIR dataset [5]. It is an academic prototype evaluated on an internal held-out test set drawn from the same dataset. It does not assess clinical deployment, and its results should not be read as clinical performance.

1.3 Objectives

The project had five objectives:

1. Build a patient-disjoint binary cataract cohort from ODIR, with auditable label, exclusion and split rules.
2. Compare a custom CNN, ResNet18 and EfficientNet-B0 under a common protocol, selecting checkpoints and the primary model on validation data only.
3. Evaluate each model at a common reference threshold (0.5) and at a model-specific validation-selected threshold that targets at least 90% validation sensitivity, and report held-out test results with patient-level bootstrap intervals.
4. Test in a matched-cohort experiment whether healthy-only or mixed negative training examples change ResNet18 behaviour.
5. Examine errors and check Grad-CAM maps with a parameter-randomization test, stating what each analysis can and cannot show.

2 Related Work

2.1 Cataract Detection from Fundus Images

Early deep learning work applied a deep convolutional network to clinical fundus images (up to 5,620 images) and reported best accuracies of 93.52% for cataract detection and 86.69% for grading [4]. Xie et al. [2] developed a three-class system (non-cataract, mild cataract, visually impaired cataract) on 8,395 images from 5,245 subjects at three centres. They compared DenseNet121, Inception-v3 and ResNet50, kept images from one person in a single data set, and evaluated the best model on two external test sets. Their error analysis attributed most non-cataract images misclassified as mild cataract (34 of 38) to dark photographs [2]. Tham et al. [3] trained a detector for visually significant cataract on retinal photographs from population-based studies (25,742 photographs overall) and reported an internal-test AUROC of 96.6%.

These studies define the positive class differently. Earlier grading work often relied on image blurriness, while [2] and [3] tie the label to visual function [2], [3]. They also report different metrics on different populations, so their numbers cannot be compared directly with a binary detector trained on ODIR labels.

2.2 The ODIR Dataset

The ODIR dataset was released as a multi-disease benchmark of fundus images from both eyes of patients, covering eight disease categories [5]. In the training data used here, annotations are given per eye as diagnostic keywords that may combine several conditions. Cataract studies built on ODIR often use subsets and different protocols. For example, one study reports 100% accuracy for two-class normal-versus-cataract classification and 96.9% for four-class grading on augmented ODIR images, with 94% on the original images [6]. Accuracy depends on class balance, and on whether augmentation and splitting kept patients separate. Direct numerical comparison with this report is therefore not meaningful.

Table 2.1 summarizes the cataract studies cited above, using the figures as stated by their authors. The studies differ in data, label definition and metric, which is why this report does not rank itself against them.

Table 2.1. Summary of cited cataract-detection studies and the ODIR dataset (figures as stated by the authors).

Study	Data	Task or label	Reported result
Zhang et al. 2017 [4]	Clinical fundus images (up to 5,620)	Cataract detection and grading	Best accuracy 93.52% (detection) and 86.69% (grading)
Xie et al. 2023 [2]	8,395 images, 5,245 subjects, three centres; two external test sets	Non-cataract, mild, visually impaired cataract	Best model DenseNet121; 34 of 38 non-cataract images read as mild attributed to dark photographs
Tham et al. 2022 [3]	25,742 retinal photographs from population-based studies	Visually significant cataract	Internal-test AUROC 96.6%
Li et al. 2021 [5]	Fundus images from both eyes of patients, eight disease categories	ODIR multi-disease benchmark	Dataset and benchmark baselines
Ali et al. 2022 [6]	ODIR-derived subsets, augmented and original images	Normal vs cataract; four-class grading	100% (two classes); 96.9% (four classes, augmented) and 94% (original images)

2.3 Architectures and Transfer Learning

ResNet introduced residual connections that make deep networks easier to optimize [7]. EfficientNet scales depth, width and resolution together with a compound coefficient [8]. Initializing from ImageNet weights and fine-tuning is the usual starting point in medical imaging, although a study of two large tasks, including diabetic retinopathy from fundus photographs, found that transfer gave little performance benefit and mainly sped up convergence [9]. This matters when interpreting a gap between pretrained and from-scratch models. Training here uses AdamW, which decouples weight decay from the gradient update [10].

2.4 Evaluation under Class Imbalance and Uncertainty

On imbalanced data, ROC plots can look favourable while the fraction of positive predictions that are correct stays low, so precision–recall measures are more informative [11]. Percentile bootstrap intervals [12] describe sampling variability of a metric for a fixed model. When images are clustered within patients, resampling whole patients is a cluster bootstrap [13]. Diagnostic accuracy also varies with the patient mix, a spectrum effect [14], which is relevant to the choice of negative examples.

2.5 Explainability and Sanity Checks

Grad-CAM weights the final convolutional feature maps by gradients of a target score to produce a coarse map [15]. Adebayo et al. showed that visual plausibility alone can mislead, and proposed randomization tests in which a map computed from a trained model is compared with one from a model whose parameters are randomized. In their experiments Grad-CAM

passed these checks [16]. Randomization tests therefore test sensitivity to model parameters. They do not show that maps localize pathology.

2.6 Positioning of This Work

This project adds a transparent, auditable evidence trail on one public dataset. It combines patient-level splitting, two clearly separated threshold policies, patient-level paired bootstrap intervals, a matched negative-cohort experiment and a randomization check, and it states the evidence limits of each.

3 Dataset and Cohorts

3.1 Source Data, Labels and Exclusions

The study used the original training images and annotation spreadsheet of the ODIR-2019 dataset (https://odir2019.grand-challenge.org/dataset/) [5]. The spreadsheet records diagnostic keywords separately for each eye, so each eye image is one example and a patient-level indicator was never copied to both eyes. All 7,000 eye records matched 7,000 distinct original image files; pre-processed copies were not used.

A **cataract** label required the exact cataract term in the eye's annotation (313 eye records before exclusions). A **non-cataract** label means only that cataract was not recorded; it is not a clinical confirmation of a normal lens. For evaluation, non-cataract eyes were divided into **healthy negatives** (normal fundus as the only term) and **other-condition negatives**.

Exclusion rules were fixed before any split was made, and every included and excluded record was saved with its reason. Table 3.1 lists the 500 excluded images, each with one recorded reason; lens dust accounts for 405. The single *excluded_altered_eye_state* entry corresponds to the silicone-oil-eye rule. The saved table gives no separate counts for the conflicting-term or missing-term rules. Cataract images fell from 313 to 287, so 26 cataract eyes were removed, but the exclusions are not broken down by class and the reasons for those 26 removals are unknown.

Table 3.1. Images excluded before splitting, by recorded reason.

Exclusion reason	Images
Lens dust	405
Refractive-media opacity without an explicit cataract term	63
Low image quality	21
Optic disk not visible	5
Anterior-segment image	2
Silicone-oil eye (recorded as *excluded_altered_eye_state*)	1
Image offset	1
No fundus image	1
Uncertain cataract annotation	1
Total	**500**

3.2 Eligible Cohort and Patient-Disjoint Splits

The eligible cohort holds 6,500 images from 3,345 patients: 287 cataract images (4.42%) and 6,213 non-cataract images. By evaluation group it holds 287 cataract images (196 patients), 2,818 healthy negatives (1,816 patients) and 3,395 other-condition negatives (2,110 patients). These patient counts overlap because the group is assigned per eye, so they must not be summed. Image widths range from 250 to 5,184 pixels and heights from 188 to 3,456 pixels. A hash of decoded pixels found no exact duplicates.

Images were split by patient (70%, 15%, 15%; seed 42), so both eyes of a patient share one set. The split was stratified by patient stratum (196 patients with a cataract eye, 1,080 healthy-only patients, 2,069 other-negative patients), saved, and reused for every model; the protocol did not allow choosing a different split on the basis of model performance. Patient, image-path and exact-pixel overlap checks all passed (Table 3.2, Figure 3.1). The test set is an internal holdout from the same dataset and contains only 41 cataract images, so one missed image changes test sensitivity by about 2.4 percentage points.

Table 3.2. Patient-disjoint splits (seed 42). Prevalence is the cataract share of images.

Split	Patients	Images	Cataract	Healthy neg.	Other neg.	Prevalence
Training	2,341	4,548	202	1,965	2,381	4.44%
Validation	502	979	44	422	513	4.49%
Test	502	973	41	431	501	4.21%
Total	**3,345**	**6,500**	**287**	**2,818**	**3,395**	**4.42%**

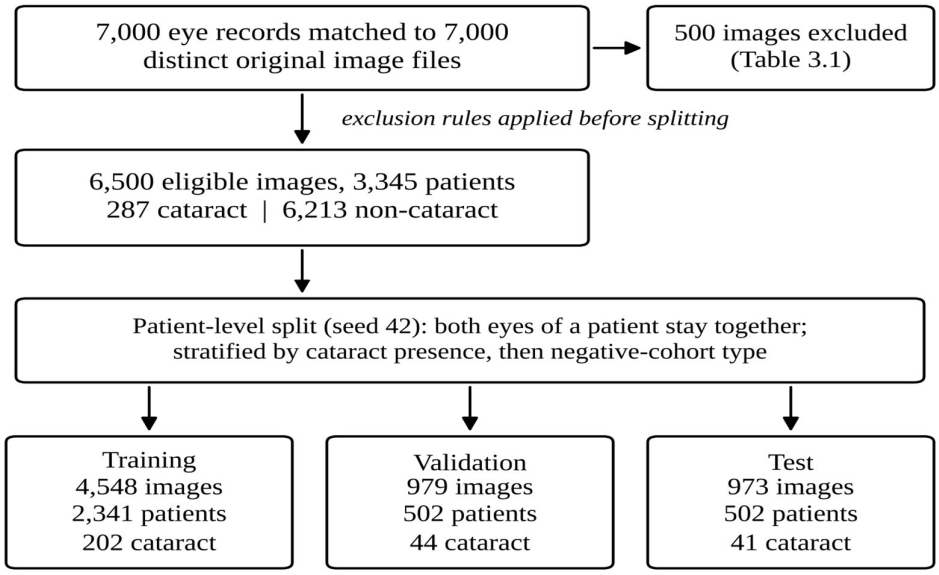


Figure 3.1. Cohort construction and patient-level split. Schematic drawn from the saved aggregate counts.

3.3 Near Duplicates and Cohort Limits

Patients were kept together, so near duplicates within a patient cannot cross splits. A difference-hash (dHash) screen across all images returned 1,893,711 candidate pairs (150 within the same patient), reflecting excessive false matches. It was rejected, and it was used neither to remove images nor to group patients. Near duplicates across different patients are therefore not ruled out. ODIR labels were not independently adjudicated here, no external dataset was used, and the effect of the 26 excluded cataract eyes on the retained case mix cannot be assessed.

4 Preprocessing

All models received the same preprocessing. Each image was read with its EXIF orientation applied, converted to RGB and resized to 224 × 224 pixels. The longer side was scaled with bicubic interpolation and the image was centred on a black padded square, so no image was stretched or cropped. Pixel values were normalized with the ImageNet channel means (0.485, 0.456, 0.406) and standard deviations (0.229, 0.224, 0.225). Training images received only mild augmentation after loading, a horizontal flip with probability 0.5 and a random rotation of up to 5 degrees. Validation and test images were not augmented.

To speed up loading, the deterministic 224 × 224 images of the training and validation sets were stored once as lossless PNG files, each checked against freshly computed pixels. On eight validation images the original and cached pipelines gave a maximum tensor difference of 0.0. A CPU benchmark on 128 images measured 16.3 images per second from the original files and 430.9 from the cache (26.4 times faster), which reflects data loading only. The cache was built after the Custom CNN was trained, so the Custom CNN used the original-image pipeline with

identical deterministic steps, while the other models trained from the cache. The notebook's loading code confirms that the Custom CNN's training and validation loaders and the final test loader all read the original files through one dataset class. Final test inference for every model used the original test images, which were never cached (Table 4.1).

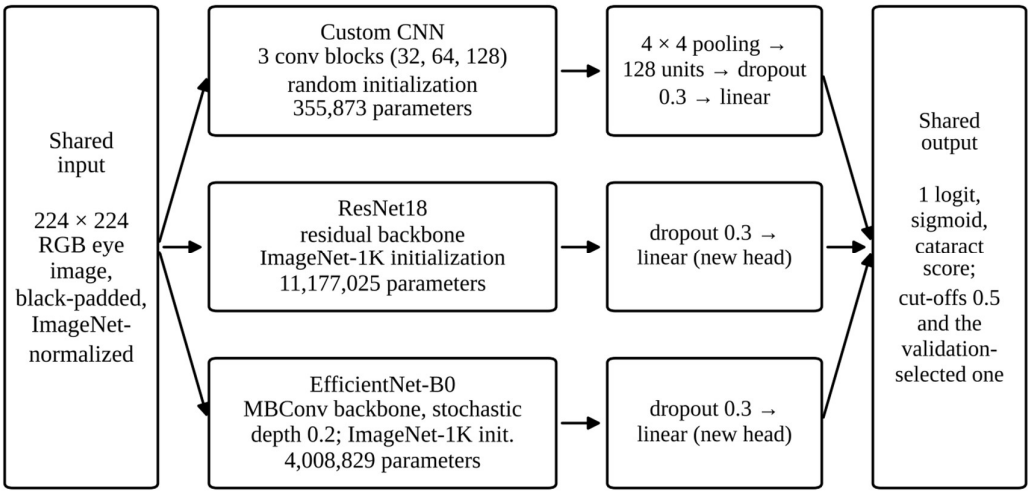
Table 4.1. Preprocessing and data-loading settings used for all models.

Setting	Value
Input	RGB, 224 × 224 pixels; EXIF orientation applied
Resizing	Aspect-preserving, centred on a black padded square; bicubic interpolation
Normalization	ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225)
Training augmentation	Horizontal flip (p = 0.5); rotation up to ±5°
Validation and test	Deterministic steps only; no augmentation
Loader	Batch size 16; no worker processes
Cache	Lossless PNG, training and validation images only; augmentation applied after loading

5 Methodology

5.1 Models

Each model maps one preprocessed image to a single logit; a sigmoid gives a cataract score between 0 and 1, and an image is predicted as cataract when its score is at or above a threshold. The scores are not calibrated probabilities. Figure 5.1 shows the three models as implemented. The **Custom CNN** is a small baseline trained from scratch (three convolution blocks with 32, 64 and 128 channels, adaptive pooling to 4 × 4, a 128-unit hidden layer with dropout 0.3). **ResNet18** [7] and **EfficientNet-B0** [8] were initialized from ImageNet-1K weights, and their classifiers were replaced by dropout (0.3) and one linear layer.



All three: same training split (4,548 images), weighted BCE (positive weight 21.51), AdamW, seed 42

Figure 5.1. The Custom CNN, ResNet18 and EfficientNet-B0 as implemented, with the shared input, the initialization of each model and the shared binary output. Original schematic drawn from the saved model configurations and notebook outputs.

5.2 Training

All three main models were trained on the same 4,548-image training set (202 cataract images) with seed 42. The loss was binary cross-entropy with logits and a positive-class weight equal to the ratio of training negatives to positives, 4,346 / 202 = 21.51. Optimization used AdamW [10] with weight decay 1×10^{-4}, batch size 16 and mixed precision. The learning rate was halved when validation AP stalled for two epochs, and the checkpoint with the highest validation AP was kept (Table 5.1).

The Custom CNN trained for up to 25 epochs at 1×10^{-3}. The two pretrained models used a three-epoch head phase (head only, 1×10^{-3}) followed by up to 22 fine-tuning epochs (backbone 1×10^{-5}, head 1×10^{-4}) with BatchNorm statistics frozen, and early stopping after five epochs without improvement. Validation loss was unweighted, so training and validation losses differ. The Custom CNN reached its epoch limit at its best validation epoch, so training length may have limited the baseline. Its training was paused and resumed three times, once repeating a partly completed epoch, and equivalence to uninterrupted training was not established.

Table 5.1. Models, training settings and selected checkpoints (validation data).

Model	Initialization	Parameters	Learning rate	Epochs completed (selected)	Validation AP
Custom CNN	Random	355,873	1×10^{-3}	25 (25)	0.6493
ResNet18	ImageNet-1K	11,177,025	Head 1×10^{-3}, then backbone 1×10^{-5} and head 1×10^{-4}	11 (6)	0.9217
EfficientNet-B0	ImageNet-1K	4,008,829	As ResNet18	10 (5)	0.9259

5.3 Model Selection and Threshold Policy

Validation AP was the selection metric for checkpoints and for the primary architecture. **EfficientNet-B0 was the primary model**, with the highest validation AP among the three main models (0.9259, against 0.9217 for ResNet18 and 0.6493 for the Custom CNN). The 0.0042 gap to ResNet18 is small on 44 validation cataract images and is not a meaningful difference. The designation was fixed before test evaluation and is not revisited after test results.

Two threshold policies are used throughout, and every threshold-dependent result carries its label. The **reference threshold** is 0.5 for every model. The **validation-selected threshold** is a prespecified, model-specific operating policy: among all distinct validation scores (plus 0, 0.5 and 1), keep thresholds giving validation sensitivity of at least 0.90, and choose the one with the highest validation specificity, breaking ties by higher sensitivity and then higher threshold. The 90% target is an illustrative research operating point, not a clinical requirement, it was not guaranteed on the test set, and thresholds were never retuned after test evaluation. Validation performance at these thresholds is selection data and not an independent estimate (Table 5.2).

Table 5.2. Frozen validation-selected thresholds and the validation operating points they produce (validation data; selection data, not independent estimates).

Model	Validation-selected threshold	Val. sensitivity	Val. specificity	TP / FP / FN / TN (validation)
Custom CNN	0.2716	0.9091	0.8556	40 / 135 / 4 / 800
ResNet18	0.7553	0.9318	0.9904	41 / 9 / 3 / 926
EfficientNet-B0	0.4883	0.9091	0.9818	40 / 17 / 4 / 918
ResNet18, healthy-only	0.8108	0.9091	0.9872	40 / 12 / 4 / 923
ResNet18, mixed-matched	0.3314	0.9091	0.9882	40 / 11 / 4 / 924

5.4 Matched Negative-Cohort Experiment

What it tests. The experiment asks whether the composition of negative training examples changes what a ResNet18 learns when the cataract examples and the number of negative images are fixed. One model saw only healthy negatives, and the other saw a mix of healthy negatives and negatives with other eye conditions, which are abnormal eyes without cataract and may be harder negatives.

Design. Each matched-cohort model used 2,167 of the 4,548 training images (47.6%), against 4,548 for each main model. Both cohorts contain the same 202 cataract images and 1,965 negative images. The healthy-only cohort uses all 1,965 healthy training negatives. The mixed cohort is a stratified random sample of the 4,346 training negatives, proportional to the negative groups: 888 healthy and 1,077 other-condition images (Table 5.3). The two models share the same ImageNet backbone, newly initialized head, preprocessing, settings, seed, positive loss weight (1,965 / 202 = 9.73, not the main models' 21.51) and the full validation set. Patient counts differ (1,365 and 1,647), and so do the training durations that early stopping allowed (10 and 25 epochs).

Table 5.3. Matched negative-cohort design: what is held constant and what differs.

Property	Healthy-only	Mixed-matched	Status
Cataract training images	202	202 (same images)	Matched
Negative training images	1,965	1,965	Matched
Healthy / other-condition negatives	1,965 / 0	888 / 1,077	Composition under test
Total training images	2,167	2,167	Matched
Initialization, settings, seed, loss weight, validation set	Identical	Identical	Matched
Training patients	1,365	1,647	Differs
Epochs completed (selected)	10 (5)	25 (20)	Differs

What the results can support. The experiment used one seed and one mixed-negative sample and was exploratory. Because patient counts and training duration differ along with composition, a difference between the models cannot be attributed to negative composition alone. At most the results describe how these two fitted models differ under each threshold policy. They cannot establish consistent improvement, identify a mechanism or quantify seed-to-seed variation, and the models should not be compared with the main ResNet18.

5.5 Implementation and Execution Record

Experiments used Python 3.13.0, PyTorch 2.7.1 (CUDA 11.8) and torchvision 0.22.1 on a laptop GPU. The notebook was executed incrementally with checkpoint recovery, and the Grad-CAM section (§20) was rerun in a separate kernel session; the corrected notebook is the source for that section and its printed results matched the earlier run. A fresh end-to-end replay

was not verified. The Custom CNN run used throughout is *custom_cnn_seed42_20261030T172651_792236Z*, identified by the later resume cell and by the frozen evaluation protocol.

6 Evaluation Protocol

6.1 Separation of Validation and Test Evidence

Validation data were used for checkpoint selection, learning-rate scheduling, early stopping, primary-model selection, threshold selection and the Grad-CAM analysis. The held-out test set was used only for final reporting with frozen checkpoints and thresholds. Validation results are always labelled as such. The test set (973 images, 502 patients, 41 cataract) was scored once per model for five frozen models, the three main models and the two matched-cohort models, at both the reference and the validation-selected threshold. Before test scoring, the first validation batch of 16 images for each model reproduced its saved scores within numerical tolerance, and the file hashes of the frozen protocol, checkpoints and test manifest were recorded; test predictions were saved with receipts that record them.

6.2 Metrics

AP (scikit-learn average precision) and ROC-AUC do not depend on a threshold. Sensitivity, specificity, precision, F1, accuracy, balanced accuracy and the confusion counts do, and each is reported with one of the two policy labels, reference (0.5) or validation-selected; precision is 0 when no image is predicted positive. False-positive rates are also reported separately for healthy and other-condition negatives. At the test prevalence of 4.21%, AP and precision are strongly prevalence-dependent, which is why a constant-score reference (AP equal to the prevalence) is drawn on precision–recall curves [11].

Table 6.1. Metrics and the threshold policy label each requires.

Metric	**Depends on threshold**	**Label required**
Average precision, ROC-AUC	No	None
Sensitivity, specificity, precision, F1	Yes	Reference (0.5) or validation-selected
Accuracy, balanced accuracy	Yes	Reference (0.5) or validation-selected
TP, FP, FN, TN counts	Yes	Reference (0.5) or validation-selected
Healthy and other-condition false-positive rates	Yes	Reference (0.5) or validation-selected

6.3 Patient-Level Bootstrap

Uncertainty on the test set was estimated with a percentile bootstrap [12] of 2,000 replicates (95% level, seed 42). In each replicate 502 patients were drawn with replacement and all of their images were included, so patients are the resampling unit [13]. Draws were not stratified by class, so the number of cataract images varies across replicates, and all 2,000 replicates were valid for every metric. The same draws were used for every model and both policies, which gives paired differences. Ten metrics were resampled: the eight listed above plus the two subgroup false-positive rates. The intervals describe sampling variability for fixed fitted models, not training-seed variability. They are pointwise with no multiple-comparison adjustment, so paired comparisons are exploratory. An interval that includes zero, or touches it, is inconclusive for that difference; it does not show equivalence.

6.4 Grad-CAM and Parameter-Randomization Protocol

Grad-CAM [15] maps were computed for the frozen EfficientNet-B0 on validation images, at its final feature block, as the rectified weighted sum of activations, normalized to 0–1 independently for each map and resized to the image. The targets were the cataract logit and its negative, giving a cataract-logit map and a negated-logit (non-cataract) map. Scores were computed in the original sequential validation batches of 16 because single-image and batched inference differed numerically; with that batch context the recomputed scores reproduced the saved scores (maximum difference 0.0).

Eight examples were chosen with a seeded rule: validation images were grouped as true positive, true negative, false positive or false negative at the validation-selected threshold 0.4883; within each group images were sorted by path, shuffled with seed 42, reduced to one image per patient, and the first two were kept. For the randomization check [16], parameters of a copy of the trained model were reset in three stages (classification head only; head and final backbone block; all layers) with seeds 42, 43 and 44. Randomized maps were compared with the trained-model maps by Pearson correlation (when both are non-constant) and mean absolute difference. No automatic pass or fail was set, and no claim of clinically correct localization is made.

7 Results

Validation and held-out test results are reported separately. Every threshold-dependent value is labelled **reference (0.5)** or **validation-selected**; AP and ROC-AUC carry no label.

7.1 Validation Results

On validation, EfficientNet-B0 had the highest AP (0.9259), followed by ResNet18 (0.9217) and the Custom CNN (0.6493); ROC-AUC was 0.9926, 0.9925 and 0.9555. At the reference threshold, validation sensitivity was 88.64% for EfficientNet-B0 (5 missed), 93.18% for ResNet18 (3 missed) and 88.64% for the Custom CNN (5 missed), with false positives of 17,

17 and 70. The selection rule used AP only, so EfficientNet-B0 was designated primary. These are selection data and not independent estimates.

7.2 Held-Out Test Results for the Main Models

All values in this section are held-out test results (973 images, 41 cataract), with 95% patient-level bootstrap intervals in parentheses. The two pretrained models ranked cataract images far better than the Custom CNN and were almost indistinguishable from each other (Table 7.1, Figure 7.1).

Table 7.1. Main models: validation AP (selection data) and held-out test AP and ROC-AUC (threshold-independent) with 95% patient-level bootstrap intervals.

Model	Validation AP	Test AP (95% CI)	Test ROC-AUC (95% CI)
Custom CNN	0.6493	0.6534 (0.4780–0.7860)	0.9451 (0.9060–0.9756)
ResNet18	0.9217	0.9149 (0.8273–0.9753)	0.9926 (0.9833–0.9987)
EfficientNet-B0 (primary)	0.9259	0.9078 (0.8105–0.9736)	0.9920 (0.9828–0.9985)

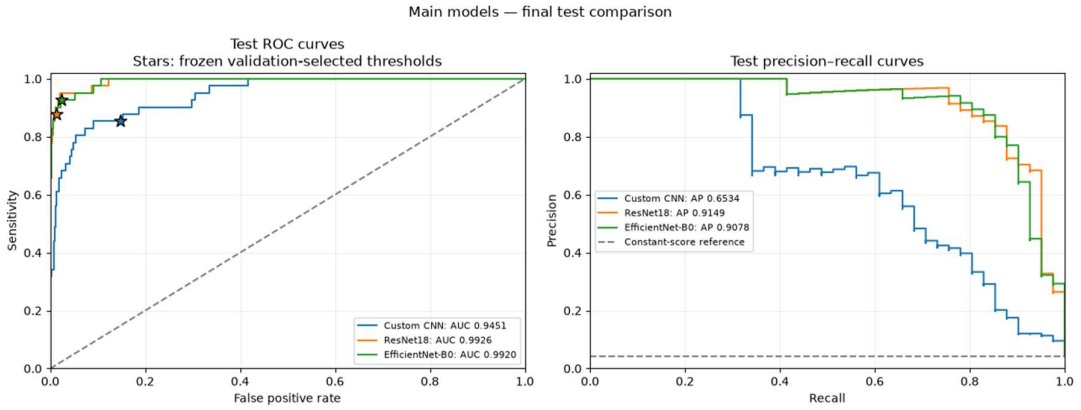


Figure 7.1. Held-out test ROC (left) and precision–recall (right) curves for the three main models. Stars mark each model's **validation-selected** threshold, not 0.5. The dashed line is the constant-score reference at the test prevalence. Corrected notebook output, §21.

At the **reference threshold (0.5)**, ResNet18 and EfficientNet-B0 each detected 38 of 41 cataract images, with 17 and 22 false positives, while the Custom CNN detected 34 and produced 75 false positives (Table 7.2). At the **validation-selected thresholds**, the 90% sensitivity target did not carry over to the test set for two of the three main models: test sensitivity was 85.37% for the Custom CNN and 87.80% for ResNet18, against validation values of 90.91% and 93.18%. Among the three main models, only EfficientNet-B0 met the target on the test set (92.68%), although its interval (83.33–100.00%) is wide. Its threshold (0.4883) gave the same test decisions as 0.5, so its two sets of results are identical. ResNet18's selected threshold (0.7553) removed four false positives (13 against 17) at the cost of two more missed images (5 against 3).

Table 7.2. Held-out test results by threshold policy, with 95% bootstrap intervals for sensitivity and specificity.

Model (threshold)	Sensitivity	Specificity	Precision	F1	FP / FN
Reference threshold 0.5					
Custom CNN	82.93% (69.23–94.23)	91.95% (89.91–93.90)	31.19%	0.4533	75 / 7
ResNet18	92.68% (83.33–100.00)	98.18% (97.22–99.04)	69.09%	0.7917	17 / 3
EfficientNet-B0	92.68% (83.33–100.00)	97.64% (96.52–98.69)	63.33%	0.7525	22 / 3
Validation-selected thresholds					
Custom CNN (0.2716)	85.37% (72.50–96.15)	85.30% (82.69–88.03)	20.35%	0.3286	137 / 6
ResNet18 (0.7553)	87.80% (74.28–97.62)	98.61% (97.75–99.35)	73.47%	0.8000	13 / 5
EfficientNet-B0 (0.4883)	92.68% (83.33–100.00)	97.64% (96.52–98.69)	63.33%	0.7525	22 / 3

Selected model versus highest observed test score. EfficientNet-B0 is the model selected on validation and remains the primary model. ResNet18 has the highest observed test AP (0.9149) and ROC-AUC (0.9926), and at the reference threshold higher specificity and precision. The paired intervals do not separate the two: the AP difference is +0.0071 (−0.0168 to 0.0366) and the ROC-AUC difference is +0.0005 (−0.0017 to 0.0033). Superiority or equivalence is therefore not established, and the higher observed test score is not used to change the selection. Both pretrained models exceeded the Custom CNN, with paired AP differences of +0.2615 (0.1343 to 0.4212) for ResNet18 and +0.2544 (0.1312 to 0.4078) for EfficientNet-B0, and ROC-AUC differences of about +0.047 whose intervals exclude zero. Among threshold-dependent differences, ResNet18 minus EfficientNet-B0 at the reference threshold differed only in the other-condition false-positive rate (−1.00 percentage points; −1.91 to −0.20). At the validation-selected thresholds ResNet18 had higher specificity and precision, but each model is then compared at its own threshold, so those differences reflect threshold choice as well as the models.

7.3 Matched Negative-Cohort Experiment

On **validation** at the reference threshold, the mixed-negative model had higher AP (0.9319 against 0.9111), fewer false positives (10 against 25) and more missed cataract images (6 against 3); ROC-AUC was 0.9922 against 0.9935. On the **held-out test set** the two models were not separated on ranking: AP was 0.9031 (healthy-only) and 0.9127 (mixed), a difference of +0.0096 (−0.0104 to 0.0316), and ROC-AUC differences were similarly inconclusive (Table 7.3).

Table 7.3. Matched-cohort ResNet18 models: held-out test results and paired differences (mixed-matched minus healthy-only) with 95% paired bootstrap intervals. pp = percentage points; FPR = false-positive rate.

Metric	Healthy-only	Mixed-matched	Difference (95% CI)
Threshold-independent			
Average precision	0.9031	0.9127	+0.0096 (−0.0104 to 0.0316)
ROC-AUC	0.9870	0.9850	−0.0020 (−0.0163 to 0.0078)
Reference threshold 0.5			
Sensitivity	92.68%	87.80%	−4.88 pp (−12.20 to 0.00)
Specificity	97.53%	98.50%	+0.97 pp (0.32 to 1.70)
Precision	62.30%	72.00%	+9.70 pp (2.56 to 17.57)
Other-condition FPR	1.80%	0.60%	−1.20 pp (−2.19 to −0.39)
Validation-selected thresholds (0.8108 | 0.3314)			
Sensitivity	87.80%	92.68%	+4.88 pp (0.00 to 12.20)
Specificity	98.28%	98.18%	−0.11 pp (−0.64 to 0.43)
Precision	69.23%	69.09%	−0.14 pp (−6.41 to 6.67)
Other-condition FPR	0.80%	1.00%	+0.20 pp (−0.62 to 1.10)

At the **reference threshold**, the mixed-negative model had higher specificity (+0.97 pp), higher precision (+9.70 pp) and a lower other-condition false-positive rate (−1.20 pp), with intervals that exclude zero. Its sensitivity point estimate was lower (−4.88 pp, two more missed images), but that interval touches zero, so the sensitivity difference is inconclusive. At the **validation-selected thresholds** the pattern changed: the sensitivity point estimate favoured the mixed model (+4.88 pp), again with an interval touching zero and so inconclusive, and the specificity, precision and false-positive differences were all inconclusive. The experiment therefore shows threshold-dependent differences in operating behaviour, not a consistent improvement, and ranking performance did not differ clearly. Differences in patient counts and training duration, one seed, one mixed sample and 14 unadjusted comparisons limit any stronger reading.

7.4 Grad-CAM Parameter-Randomization Check (Validation Images)

This check uses the eight **validation** images and the corrected notebook (§20), which reproduced the earlier outputs. Table 7.4 summarizes 24 comparisons (eight images, three seeds) per stage and target. Partial randomization changed the maps: cataract-logit correlations with the trained maps were slightly negative (−0.136 and −0.107) and scores shifted by about 0.35–0.37 on average. The negated-logit maps are less clear (−0.072 and +0.260 over 21 valid comparisons; three comparisons per stage have a constant trained map). Under full randomization every randomized map was constant, so no correlation could be computed and

the comparison is inconclusive. The protocol set no pass or fail rule, and the results do not show that maps reflect clinically correct localization.

Table 7.4. Parameter-randomization comparison for EfficientNet-B0 Grad-CAM maps (eight validation images, seeds 42–44). Pearson r compares each randomized map with the trained-model map.

Stage	Map	Valid r	Mean r	Mean abs. map diff.	Constant maps	Mean abs. score change
Head only	Cataract	24 / 24	−0.136	0.393	0	0.353
Head only	Negated	21 / 24	−0.072	0.214	1	0.353
Head + final block	Cataract	24 / 24	−0.107	0.390	0	0.368
Head + final block	Negated	21 / 24	+0.260	0.221	0	0.368
All layers	Cataract	0 / 24	undefined	0.383	24	0.360
All layers	Negated	0 / 24	undefined	0.084	24	0.360

8 Error Analysis

8.1 Scope and Evidence Limits

The per-image prediction files are not in the bundle, so analysis of **held-out test** errors relies on aggregate confusion counts (Table 8.1) and subgroup false-positive rates. Individual test errors cannot be inspected or matched across models. Section 8.4 shows Grad-CAM examples from **validation** images chosen by a seeded rule; they illustrate model behaviour and are not test errors or a representative sample of them.

8.2 Held-Out Test Errors by Model and Policy

Because cataract prevalence is about 4%, false positives outnumber missed images for every model. At the reference threshold EfficientNet-B0 made 22 false-positive and 3 false-negative errors, so 22 of its 60 positive calls (37%) were wrong, against 17 and 3 for ResNet18. The Custom CNN missed 7 images and made 75 false positives, and its validation-selected threshold raised false positives to 137 while recovering one image. Raising a threshold trades false positives for misses: ResNet18 at 0.7553 and the healthy-only model at 0.8108 each missed 5 images, against 3 at 0.5. A difference of one or two missed images moves sensitivity by 2.4 to 4.9 percentage points, and without per-image predictions it is not possible to tell which images were missed or whether different models missed the same ones.

Table 8.1. Held-out test confusion counts by model and threshold policy (973 images; 41 cataract, 932 non-cataract).

Model (threshold)	TP	FP	FN	TN
Reference threshold 0.5				
Custom CNN	34	75	7	857
ResNet18	38	17	3	915
EfficientNet-B0	38	22	3	910
ResNet18, healthy-only	38	23	3	909
ResNet18, mixed-matched	36	14	5	918
Validation-selected thresholds				
Custom CNN (0.2716)	35	137	6	795
ResNet18 (0.7553)	36	13	5	919
EfficientNet-B0 (0.4883)	38	22	3	910
ResNet18, healthy-only (0.8108)	36	16	5	916
ResNet18, mixed-matched (0.3314)	38	17	3	915

8.3 False Positives by Negative Subgroup (Held-Out Test)

Table 8.2 splits test false positives by the evaluation group of the negative eye (431 healthy and 501 other-condition negatives). For the three main models, under both policies, healthy negatives had higher false-positive rates than other-condition negatives. At the reference threshold ResNet18 produced false positives on 15 of 431 healthy images (3.5%) but 2 of 501 other-condition images (0.4%), and EfficientNet-B0 on 15 (3.5%) and 7 (1.4%); the Custom CNN showed the same ordering at higher rates (11.1% and 5.4%). This pattern is observed, and possible causes, such as differences in image quality, acquisition conditions or annotation practice between groups, were not tested. The subgroup intervals are wide, for example 0.0% to 1.0% for ResNet18 on other-condition negatives.

Table 8.2. Held-out test false positives by negative subgroup: count (rate). Healthy negatives n = 431; other-condition negatives n = 501.

Model	Healthy, reference (0.5)	Other-condition, reference (0.5)	Healthy, validation-selected	Other-condition, validation-selected
Custom CNN	48 (11.1%)	27 (5.4%)	83 (19.3%)	54 (10.8%)
ResNet18	15 (3.5%)	2 (0.4%)	11 (2.6%)	2 (0.4%)
EfficientNet-B0	15 (3.5%)	7 (1.4%)	15 (3.5%)	7 (1.4%)
ResNet18, healthy-only	14 (3.2%)	9 (1.8%)	12 (2.8%)	4 (0.8%)
ResNet18, mixed-matched	11 (2.6%)	3 (0.6%)	12 (2.8%)	5 (1.0%)

8.4 Validation Grad-CAM Examples

Table 8.3 lists the eight validation images used for Grad-CAM (chosen at the validation-selected threshold 0.4883), and Figure 8.1 shows one of each outcome. They are **validation images**, not held-out test errors. The observations below are visual impressions, not clinical assessments.

Table 8.3. Validation images used for Grad-CAM (EfficientNet-B0; outcome at the validation-selected threshold 0.4883).

Row	Evaluation group	Outcome	Validation score	In Figure 8.1
719	Cataract	True positive	0.9924	No
229	Cataract	True positive	0.8453	Yes
458	Other-condition negative	True negative	0.0007	No
647	Healthy negative	True negative	0.0028	Yes
652	Other-condition negative	False positive	0.7639	Yes
654	Other-condition negative	False positive	0.7768	No
664	Cataract	False negative	0.1398	No
724	Cataract	False negative	0.3543	Yes

False positives. Both validation false positives (scores 0.7639 and 0.7768) are other-condition negatives and look darker than typical images, with a bright optic disk. Their cataract-logit maps highlight regions beside the optic disk with sharp block-shaped edges. **False negatives.** The two missed cataract images (0.1398 and 0.3543, below 0.4883) differ from each other: row 664 is well exposed with a bright rim reflection and a diffuse map, and row 724 is hazy and low in contrast with a narrow band of high values below the centre; the second score is a borderline miss.

True negatives. In both true-negative examples (rows 458 and 647) the highest values of the cataract-logit map fall on the black padding outside the fundus disc, as shown for row 647 in

Figure 8.1. This is an observation in these validation examples and its cause was not investigated. Independent normalization changes each map's colour scale, so colours are not comparable across maps, but it does not explain where the peak lies.

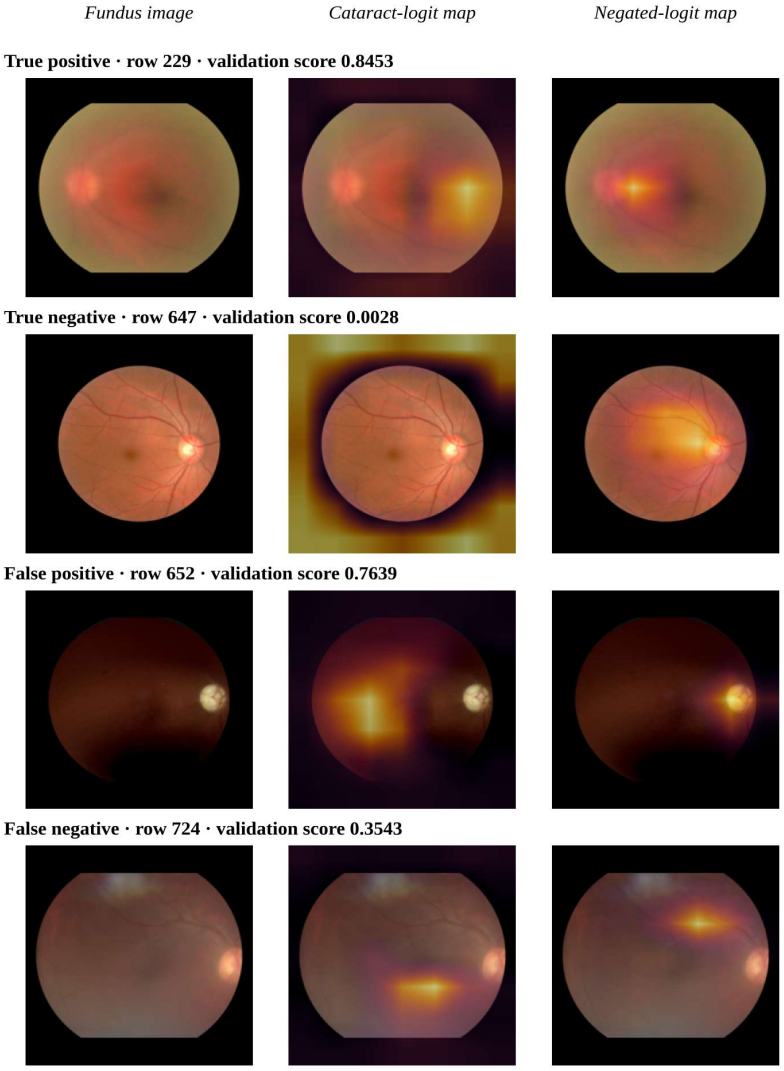


Figure 8.1. Selected **validation** Grad-CAM examples for EfficientNet-B0 at the validation-selected threshold 0.4883: one example for each outcome, showing the fundus image, the cataract-logit map and the negated-logit map. Tiles are cropped from the corrected notebook's panels with labels re-typeset; each map is independently normalized. These are not held-out test errors.

8.5 Summary of Error Patterns

False positives dominate the errors at every threshold, and changing a threshold moves errors between types without removing them. For the main models healthy negatives were misclassified more often than other-condition negatives, a pattern the aggregate data cannot explain. The weaker Custom CNN produced many more false positives than the pretrained

models. These findings are limited by the missing per-image test predictions, the 41 test cataract images, and the use of eight validation images for qualitative review.

9 Discussion

9.1 Main Findings

The clearest result is the gap between the two ImageNet-pretrained models and the Custom CNN: held-out test AP was 0.915 and 0.908 against 0.653, with paired intervals that exclude zero. Part of that gap may reflect training budget and not architecture alone. The Custom CNN reached its epoch limit at its best validation epoch, and a study of medical imaging found that ImageNet pretraining often mainly speeds convergence [9]. The two pretrained models are not separated. ResNet18's higher observed test AP and ROC-AUC lie within sampling variability (paired AP difference +0.0071, interval −0.0168 to 0.0366). The validation rule chose EfficientNet-B0 on a 0.0042 AP margin, so the primary-model choice follows a prespecified rule and is not evidence of superiority. With 41 test cataract images, ranking differences of this size cannot be resolved. The test ROC-AUC values near 0.99 should not be read against the 96.6% internal AUROC reported for a population-based cohort [3]: the labels, prevalence and cohorts differ, and this report's test set is an internal holdout.

9.2 Operating Points

The validation-selected thresholds came from only 44 validation cataract images, where one image moves sensitivity by 2.3 percentage points. Two of the three main models fell short of the 90% target on the test set (85.4% and 87.8%). EfficientNet-B0 met it (92.68%), but its interval (83.33–100.00%) does not establish that the target holds more generally, and its threshold gave the same test decisions as 0.5. At a prevalence near 4%, even good specificity leaves a visible false-positive burden: 22 of EfficientNet-B0's 60 positive calls were wrong at 0.5. Precision–recall views show this more directly than ROC curves [11]. A threshold tied to a sensitivity target is a policy choice that needs more validation positives, or interval-aware selection, before it can be trusted.

9.3 Negative-Cohort Composition

The matched-cohort experiment did not show that mixed negatives improve ranking. Threshold-dependent differences changed with the operating policy, and the sensitivity intervals touched zero, so those differences are inconclusive. The experiment does illustrate that operating-point metrics can depend on the composition of the training negatives and on threshold choice, which is consistent with the broader point that diagnostic performance depends on case mix [14]. Because patient counts and training duration also differed, composition is only one candidate explanation. Replication across seeds and mixed samples, with matched patient counts and epochs, would be needed to say more.

9.4 Error Patterns and Explainability

For the three main models, healthy negatives were misclassified more often than other-condition negatives. This was unexpected and remains an observation, since image quality, acquisition and annotation differences were not tested. In an external study, dark photographs accounted for most non-cataract images read as mild cataract [2]. The two validation false positives examined here also look dark, but two images cannot show that darkness explains the model's errors. Partial parameter randomization changed the Grad-CAM maps, so the maps depend on learned parameters, but full randomization gave constant maps and no valid correlation, so the check is inconclusive [16]. In the validation true-negative examples the cataract-logit maps peaked on the black padding, which shows that map intensity alone should not be read as lesion localization.

9.5 Implications

The results support patient-level splitting, reporting both a fixed and a validation-selected threshold, and attaching intervals to comparisons. They do not support a clinical claim: the test set is an internal holdout from one dataset, near duplicates across patients were not cleared, and labels were not independently adjudicated. The next steps are external validation, several training seeds, per-image error review and an audit of why 26 cataract eyes were excluded.

10 Limitations

Table 10.1 lists the evidence limits that apply to the findings above.

Table 10.1. Evidence limits and their consequences.

Evidence limit	Consequence
ODIR labels were not independently adjudicated; non-cataract means no cataract recorded; reasons for the 26 excluded cataract eyes are unavailable.	Label noise and selection effects on the retained cases cannot be assessed.
The test set holds 41 cataract images from one dataset (internal holdout).	Wide intervals; one image moves sensitivity by 2.4 points; no external validity claim.
Near duplicates across patients were not cleared.	Test performance may be optimistic.
One training seed per model; the bootstrap covers fixed fitted models only.	Training-seed variability is not measured.
The Custom CNN was paused and resumed three times and reached its epoch limit at its best epoch.	Its gap to the pretrained models may partly reflect training budget.
Matched cohorts differ in patient count and training duration; one seed and one mixed sample.	Differences cannot be attributed to composition alone; results are exploratory.
The 90% sensitivity target used 44 validation positives; scores are not calibrated probabilities.	Operating points are unstable and are not clinical thresholds.
Per-image prediction files, checkpoints and saved cohort manifests are not in the bundle (hashes only).	Results rest on aggregate files; per-image error review and recomputation are not possible.
Grad-CAM covers eight validation images; fully randomized maps were constant.	The randomization check is inconclusive and illustrative.
The notebook was executed incrementally; Section 20 was rerun in a separate kernel; Run All was not verified.	Reproduction needs the saved artifacts and the frozen protocol.
Clinical use was not assessed.	This is an academic prototype.

11 Conclusion

This project built a patient-disjoint binary cataract cohort from ODIR (6,500 images, 3,345 patients, 287 cataract images) and compared three models under one protocol, with checkpoints, the primary model and operating thresholds all chosen on validation data. Both ImageNet-pretrained models ranked cataract images far better than the custom CNN on the held-out test set. EfficientNet-B0, selected on validation, is the primary model, while ResNet18 had the highest observed test scores; their difference is inconclusive. At the reference threshold both pretrained models detected 38 of 41 cataract images, and among the three main models only EfficientNet-B0 met the 90% validation-selected sensitivity target on the test set.

The matched-cohort experiment showed threshold-dependent differences but no clear ranking change between healthy-only and mixed negatives. Error analysis showed that false positives dominate, with healthy negatives misclassified more often than other-condition negatives for the main models, a pattern whose causes are untested. The Grad-CAM randomization check was inconclusive. Future work should add external validation, several training seeds, calibrated

scores, per-image error review and a label audit before any claim beyond this internal comparison.

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Appendix A Reproducibility Summary

Table A.1. Key settings needed to reproduce the reported results.

Item	Setting
Environment	Python 3.13.0; PyTorch 2.7.1+cu118; torchvision 0.22.1; laptop GPU
Data and splits	ODIR original training images; 6,500 images, 3,345 patients; patient-level 70/15/15 split, seed 42
Preprocessing	224 × 224, aspect-preserving black padding, bicubic; ImageNet normalization; training flip p = 0.5 and rotation ±5°
Training	AdamW (weight decay 1×10^{-4}); batch 16; weighted BCE (21.51; 9.73 for matched cohorts); mixed precision; seed 42
Selection	Highest validation AP; early stopping patience 5; learning rate halved after 2 stalled epochs
Thresholds	Reference 0.5; validation-selected (sensitivity ≥ 0.90): 0.2716, 0.7553, 0.4883 (main models); 0.8108, 0.3314 (matched cohorts)
Bootstrap	2,000 patient-level paired replicates; 95% percentile intervals; seed 42
Grad-CAM	EfficientNet-B0, final feature block; eight validation images by a seeded rule; randomization seeds 42–44
Custom CNN run	custom_cnn_seed42_20261030T172651_792236Z (resume cell and frozen protocol)
Execution	Notebook executed incrementally with checkpoint recovery; Section 20 rerun in a separate kernel; Run All not verified

Artifacts. The project bundle contains the configurations, training summaries, split and audit files, frozen evaluation protocol, aggregate test results, confidence intervals, subgroup metrics, Grad-CAM tables and figures, and the corrected notebook. It does not contain the raw images, model checkpoints (identified by recorded hashes), saved cohort manifests, the near-duplicate candidate file, or the per-image validation and test prediction files. Reproduction needs the ODIR images in their original layout with the saved manifests, checkpoints, configurations and frozen protocol. Table A.2 lists the primary evidence sources by chapter; a line-by-line trace accompanies this report as the master evidence draft.

Table A.2. Primary evidence sources by chapter (paths relative to the project bundle).

Chapter	Primary sources
3 Dataset and cohorts	*data/splits/*; *results/data_audit/*; corrected notebook §§4–8
4 Preprocessing	*results/data_audit/preprocessing_protocol.json*; *results/cache_benchmark/*; notebook §13
5 Methodology	*config.json* and *training_summary.json* per model; *results/model_comparison/frozen_evaluation_protocol.json*; *cohort_summary.csv*; *experiment_protocol.json*
6 Evaluation protocol	*frozen_evaluation_protocol.json*; *sanity_protocol.json*; notebook §§19–22
7 Results	*results/final_test/* (metrics, confidence intervals, paired differences); *validation_comparison/*; *randomization_summary.csv*
8 Error analysis	*test_subgroup_metrics.csv*; *gradcam_example_results.csv*; corrected notebook §20 figures